

Automatic evaluation and difficulty adjustment system for skiing games

Overview

This system is a ski game evaluation and rehabilitation training platform specially designed for elderly users. In addition to providing entertainment experience, the system collects and analyzes users' physical condition and reaction ability data, customizes personalized game difficulty for elderly users, and helps them safely and effectively carry out rehabilitation training. Using advanced 3D engine rendering technology, the system provides users with an immersive gaming experience, while combining large model technology to improve the accuracy of evaluation and the personalization of suggestions.

Technical architecture

Architecture Overview

The project is divided into two main parts.

1. **Ski game** : 3D ski game developed using THREEJS.
2. **Automatic difficulty adjustment system** : A system developed using LangChain and OpenAI API that dynamically adjusts game difficulty and game recommendations based on user performance.

Functional Button Introduction

Game ability assessment

Before starting the game, users need to fill out a detailed ability assessment form so that the system can provide users with a customized gaming experience based on this data. For the needs of elderly users, the assessment form includes the following key indicators.

- **Age** : used for basic user classification, and adjust the evaluation criteria to make it more in line with the physiological characteristics of the elderly.
- **Gender** : Help the system better understand the physiological differences of users and provide more suitable suggestions.
- **Single leg standing time** : Evaluate the user's lower limb strength and balance ability, especially suitable for evaluating the balance function of the elderly.

- **Number of standing-sitting times within 30 seconds** : Measure the user's muscle endurance and leg strength, which is an important indicator to evaluate the lower limb function of the elderly.
- **Pace** : Evaluate the user's movement speed, pace is an important factor in measuring the elderly's mobility.
- **Reaction time** : Used to evaluate the user's neural response ability, the evaluation of reaction time can help identify potential problems in the nervous system.
- **Gait stability score** : Through the stability score of different terrains such as flat ground, slope, and grassland, the user's gait stability is comprehensively evaluated, which is particularly important for fall prevention training.
- **Correct Response Rate (Correct Response Rate)** : A measure of the proportion of times a user responds correctly during a game (e.g., the ratio of the number of times an obstacle is avoided to the total number of responses).
- **Average Reaction Time (Average Reaction Time)** : Records the average reaction time of the user in the game to help evaluate the user's reaction speed and nervous system status.
- **Max Reaction Time (Max Reaction Time)** : Record the slowest time the user reacts and evaluate the user's performance in extreme situations.
- **Minimum Reaction Time (Min Reaction Time)** : Record the user's fastest reaction time and evaluate the user's best performance.
- **Left Leg Strength** : Measure the strength of the user's left leg to help assess the muscle balance of the left and right legs.
- **Right Leg Strength** : Measure the strength of the user's right leg to help assess the muscle balance of the left and right legs.
- **Leg Strength Symmetry (Leg Strength Symmetry)** : Calculate the symmetry of left and right leg strength to assess whether the user has significant muscle imbalance issues, which is important for preventing falls and improving gait.

Game difficulty adjustment module

Through large-scale model technology, the system can conduct deeper analysis of user-provided data. Large-scale models can not only understand the user's current health status, but also predict the user's future performance based on historical data and the performance of other users, thereby intelligently adjusting the overall difficulty of the game.

- **Overall Difficulty** : By integrating various ability parameters, including newly added advanced indicators, the system automatically adjusts the overall difficulty of the game to ensure that the game is both challenging and does not exceed the user's ability range.

- **Number of obstacles** : Adjust the number and distribution of obstacles in the game according to the user's reaction time, correct response rate, and leg strength symmetry to prevent fatigue or injury caused by excessive challenges.
- **Skiing speed** : The system will adjust the skier's speed according to the user's average reaction time and maximum reaction time, making the game both safe and fun.
- **Balance challenge** : According to the user's gait stability and leg strength symmetry, adjust the balance difficulty in the game, which helps to improve the elderly's balance ability and reduce the risk of falling.

3D engine rendering function

This system uses advanced 3D engines (such as Three.js and Canon.js) for game scene rendering, providing users with a realistic skiing experience.

- **Immersive experience** : Through 3D engine rendering, the game scenes are more realistic, and elderly users can experience the ski resort scene as if they were there. This immersive experience not only increases the fun of the game, but also improves user engagement.
- **Dynamic light and shadow effects** : The 3D engine supports dynamic light and shadow effects, making the snow trails, obstacles and other elements in the game more three-dimensional and realistic, improving the visual experience.
- **Physics Engine Simulation** : Using physics engines such as Canon.js, the game realistically simulates the movement, collision, and gravity of skiers, making the game operation more intuitive and natural, especially suitable for rehabilitation training for elderly users.

Analysis and Recommendation Module

After the game is over, the system will provide a detailed analysis report to help elderly users better understand their performance and use large-scale model technology to obtain further rehabilitation suggestions.

- **Game performance analysis** : including game duration, score, number of obstacles avoided, etc., to help users understand their game performance.
- **Personal information analysis** : combined with the user's basic information and game performance, provide targeted evaluation to help users understand their own physical condition.
- **Game suggestion** : Based on game performance, provide suggestions for further improving game skills, and encourage elderly users to continue moderate challenges.
- **Rehabilitation advice** : For the rehabilitation needs of elderly users, provide personalized rehabilitation advice to help users gradually recover their physical functions through games.

- **Body parts that need to be strengthened** : Combining game performance and new advanced indicators, the system will point out the body parts that need to be further strengthened and provide relevant training suggestions.

System advantages

Personalized difficulty adjustment designed specifically for elderly users

The system specifically targets the physiological characteristics of elderly users, customizes the difficulty of the game to ensure that the game can provide moderate challenges without causing excessive burden on their bodies. The newly added advanced indicators make the evaluation more comprehensive and scientific, and can effectively help the elderly with rehabilitation training.

Comprehensive rehabilitation assessment and personalized advice

Combined with large-scale model technology, the system can accurately evaluate the physical condition of elderly users and provide comprehensive rehabilitation suggestions. Through the analysis of newly added indicators, the system can discover potential health risks of users (such as muscle asymmetry, slow reaction, etc.), and provide personalized training suggestions to help elderly users gradually recover their physical functions.

Immersive 3D gaming experience

The game uses an advanced 3D engine for rendering, providing realistic ski scenery and natural sports simulations, allowing elderly users to immerse themselves in the game. Through this immersive experience, the game not only increases entertainment, but also improves user participation in sports, which helps with rehabilitation training.

Data structure



User dimension:

- `user_id` : User Unique Device Identifier
- `Age` : User age
- `Gender` : User gender
- `physical_condition` : General physical condition of the user
- `balance_ability` : User balance ability
- `reaction_speed` : User response speed
- `lower_limb_strength` : User lower limb strength
- `exercise_frequency` : User frequency of movement
- `Height` : user height
- `Weight` : user weight
- `BMI` : user's body mass index
- `Vision` : the user's vision condition
- `Hearing` : The user's hearing condition
- `cognitive_function` : Cognitive Function Status of Users

Game dimension:

- `game_id` : Unique Device Identifier for game sessions
- `Difficulty` : Game difficulty level
- `game_type` : Type of game (e.g. reaction time test, leg strength test, etc.)
- `game_duration` : Duration of the game
- `response_type` : Type of user response (e.g. keystrokes, pedals, etc.)

Game result dimension:

- `correct_response_rate` : Correct response rate
- `avg_reaction_time` : Average reaction time
- `max_reaction_time` : Maximum reaction time
- `min_reaction_time` : Minimum reaction time
- `left_leg_strength` : Left leg strength test results

- `right_leg_strength` : Right leg strength test results
- `leg_strength_symmetry` : Symmetry of leg strength
- `Score` : Game Score

Test data

1. User data



- Age: 72 years old
- Gender: Female
- Evaluation indicators:
 - Standing time on one leg: 12 seconds
 - Stand up and sit down within 30 seconds: 15 times
 - Walking speed: 3 meters/second
 - Response time: 300 milliseconds
 - Gait stability score:
 - Flat ground: 8 points
 - Slope: 6 minutes
 - Grassland: 5 points
 - Correct response rate: 0.6
 - Average reaction time: 2 seconds
 - Maximum reaction time: 3 seconds
 - Minimum reaction time: 1.5 seconds
 - Left leg strength: 30kg
 - Right leg strength: 35kg
 - Leg strength symmetry: 0.86

2. Game difficulty adjustment



Based on your evaluation results, we have customized the following game difficulty settings for you:

Overall difficulty

Difficulty level: Medium

Considering your age (72 years old), gender, and various physical indicators, we have set a moderate overall difficulty for you. As you progress, the difficulty of the game will gradually increase.

Specific adjustment items

1. Number of obstacles

Difficulty level: Medium

We adjusted the number and distribution of obstacles based on your single-leg standing time (30 seconds) and standing-sitting frequency (15 times). This will help you gradually improve your lower limb strength and coordination.

2. Skiing speed

Difficulty level: Medium

Considering your walking speed (3 meters/second) and reaction time (300 milliseconds), we have adjusted the skiing speed accordingly. This will help you gradually improve your reaction speed and coordination while being safe.

3. Balancing challenges

Difficulty level: Medium

Based on your gait stability performance on different terrains, we have adjusted the difficulty of the balance challenge in the game. This will help you gradually improve your balance ability during the game.

3. Results analysis



Game performance analysis

1. **Reaction speed:** Your average reaction time is 2 seconds, which is relatively normal for a 72-year-old. Although it may pose a challenge to your game performance, keep practicing and you will see continuous improvement!
2. **Correct Response Rate:** Your correct response rate is 60%, indicating that there is room for improvement in your reaction accuracy in the game. With continuous practice and adjustment, you will definitely be able to further improve this indicator!
3. **Number of collisions:** You have had a total of 3 collisions in the game, indicating that you may feel stressed when multitasking. Don't be discouraged, as your reaction speed gradually increases, these collisions will decrease!
4. **Total score and collecting coins:** Your total score is 850 points, and you have collected 15 coins, which is already a good result! Keep working hard, and as your skills improve, your score will continue to increase!

Overall, your game performance is good. Keep up the good work and believe that you will perform better in future challenges!

Personal information analysis

1. **Leg strength symmetry:** Your left leg strength is 30kg and your right leg strength is 35kg, with a symmetry of 0.86. Although there is some asymmetry, with targeted training, you will be able to significantly improve this.
2. **Gait stability:** On flat ground, your gait stability score is 8, which is quite good! Although the score is lower on slopes and grass, you already have a good foundation, and with further training, you will definitely make progress on these terrains.
3. **Reaction time:** An average reaction time of 300 milliseconds, while common in the elderly population, can still affect your daily activities and gaming performance. Keep practicing reaction speed and you will gradually see improvements and reduce the risk of falls.

Your physical condition has a good foundation, continue to maintain a positive attitude towards exercise, and you will definitely see more progress!

Game suggestions

1. **Reduce Skiing Speed:** Reduce the pressure of reaction time and reduce collisions. Reducing speed will extend your time window to deal with obstacles, thereby improving game safety. Keep going and you will gradually find that your reaction ability is improving!
 2. **Reduce the number of obstacles:** Reduce the number of obstacles in the game and gradually increase the difficulty of the challenge, which can gradually improve your neuromuscular coordination and reaction ability. Steady progress, challenge yourself, and you will make greater progress in the game!
 3. **Gradually increase the difficulty of the game:** To improve reaction speed and correct reaction rate, it is recommended that you gradually increase the difficulty of the game. This will help you adapt to higher intensity neural stimulation and enhance your overall game performance. Believe in yourself, challenging higher difficulty will bring more sense of achievement!
- These adjustments will help you gradually improve your game performance. Keep working hard and you will definitely make satisfactory progress!

Rehabilitation advice

1. **Strengthen the quadriceps and hamstrings of the left leg:** Gradually strengthen these muscle groups through resistance training (such as using elastic bands or light weight strength training). With consistent training, your leg strength will gradually balance out, helping you walk more confidently in daily life.
 2. **Increase gait stability training:** Especially training on uneven terrain such as slopes and grasslands. It is recommended to use balance boards or cushions for simulated terrain training. These exercises can enhance the coordination of your vestibular system and proprioception system, thereby improving gait stability on different terrains. Each training will make you more robust on different terrains!
 3. **Conduct reaction time and hand-eye coordination training:** Such as using fast-paced reaction games or hand-eye coordination exercises to enhance the reaction speed of the nervous system. These exercises will help improve your daily coping ability and reduce the possibility of collisions in the game. Keep practicing and your reaction speed will definitely improve!
- Through these rehabilitation trainings, you will gradually enhance your physical fitness and coordination, continue to work hard, and your health will continue to improve!

Body parts that need to be strengthened

1. **The quadriceps and hamstrings of the left leg:** These muscles are essential for maintaining knee stability and gait symmetry. By strengthening the muscles in these areas, you can significantly improve your balance when walking and standing, and reduce the risk of falling. With regular exercise, you will feel more stable in your daily activities!
 2. **Gait stability:** Especially for gait stability training on slopes and grass, it is recommended to use a balance board or cushion for training. This will enhance the reaction ability and coordination of the lower limb muscles, prevent falls, and improve your ability to cope with different terrains. Every exercise will be an important step towards a more stable gait!
 3. **Hand-Eye Coordination:** Improve reaction speed and reduce the risk of collisions in the game through specialized hand-eye coordination training (such as using reaction balls or fast-moving target training). These exercises will help you respond faster to dynamic changes in the game, enhance overall safety and confidence. Face every challenge with confidence and you will continue to improve!
- As long as you persist in exercising and training, you will see significant improvements in every important part of your body. Keep a positive attitude, and your efforts will definitely pay off!

4. Appendix (Large Model Prompt Words):



Based on the background information and game data of the following users, generate a comprehensive analysis and recommendation text. Please ensure that the content has the following characteristics:

1. **** Highly professional ****: Analysis and recommendations should be based on the latest sports science and medical science research results, using professional terminology, and providing scientific basis (rationale) for each recommendation. Ensure professional language while maintaining easy understanding.
2. **** Highly personalized ****: All analysis and recommendations must be tailored to the user's specific background information, such as age, gender, physical assessment data, and game performance. The content should reflect a deep understanding of the user's unique needs.
3. **** High operability ****: All suggestions must be specific and practical, and users can directly apply them to daily life or game training. Suggestions should include clear steps or action plans, and explain the purpose and expected results of each step.
4. **** Ordered sequence ****: Please present the analysis and recommendations of each module in an ordered sequence (i.e. a list of numbers) to improve readability and execution.

User background information:

- Age: 72 years old
- Gender: Female

Evaluation indicators:

Standing time on one leg: 12 seconds

- Stand up within 30 seconds - Sit down 15 times
- Walking speed: 0.8 meters/second
- Response time: 300 milliseconds

Gait stability score:

- Flat ground: 8 points
- Slope: 6 points
- Grassland: 5 points
- Correct response rate: 0.6

Average reaction time: 2 seconds

Maximum reaction time: 3 seconds

Minimum reaction time: 1.5 seconds

- Left leg strength: 30kg

- Right leg strength: 35kg

- Leg strength symmetry: 0.86

Game data:

- Total score: 850 points

- Game duration: 5 minutes

Number of obstacles avoided: 20

Number of collected coins: 15

Number of collisions: 3 times

Requirements for generating content:

1. ** Game performance analysis **:

Analyze user performance in the game using ordered sequences, including reaction speed, collision frequency, score, etc.

The analysis should identify the user's strengths and weaknesses, and explain these performances in conjunction with sports science theory.

Please provide a detailed explanation of the reasons for each performance and its impact on the overall gaming experience.

2. ** personal information analysis **:

Analyze user health assessment data using ordered sequences, including gait stability, leg strength symmetry, reaction time, etc.

Explain how health data affects users' daily life and gaming performance.

Provide in-depth interpretation of user health status and propose possible health risks or room for improvement.

3. ** Game suggestion **:

Provide specific game adjustment suggestions based on game performance and personal information.

It is recommended to consider the user's age, reaction time, and strength symmetry to help users gradually improve their gaming performance.

Please ensure that the suggestions are specific, practical, and can be directly applied by users in the game. Use ordered sequence to present suggestions.

4. ** Rehabilitation advice **:

Propose personalized rehabilitation training suggestions, combined with medical science, to help users enhance lower limb strength, improve balance, and improve

reaction speed.

Each recommendation should be accompanied by scientific evidence and an explanation of why this rehabilitation program is suitable for the user's specific situation.

Please provide specific training steps or exercises and explain the expected results of each step. Present suggestions using ordered sequences.

5. Body parts that require special attention:

Point out the body parts that users need to pay special attention to and strengthen their training, and explain why these parts are crucial for overall health and gaming performance.

Provide specific training suggestions and steps to help users effectively improve the function of these areas. Present suggestions in an ordered sequence.

Please integrate the above analysis and suggestions into a coherent and detailed text, ensuring that the content is professional, personalized, and highly actionable.